

## 3.3 Worksheet

Determine whether the following series converge or diverge. Use any test or idea that is helpful from any section from 3.1 – 3.3.

1.  $\sum_{k=1}^{\infty} \frac{7}{3k}$

2.  $\sum_{k=1}^{\infty} \frac{4k^3 + 2k}{2k^3 - 1}$

3.  $\sum_{n=1}^{\infty} \arctan(n)$

4.  $\sum_{n=1}^{\infty} \frac{\pi^k}{3^{2k+1}}$

$$5. \sum_{n=1}^{\infty} \sqrt[n]{2}$$

$$6. \sum_{n=1}^{\infty} \frac{1+2^n}{3^n}$$

$$7. \sum_{n=1}^{\infty} \frac{n-1}{3n-1}$$

$$8. \sum_{n=4}^{\infty} \frac{1}{n-1}$$

9.  $\sum_{n=1}^{\infty} \left( \frac{1}{e^n} + \frac{1}{n} - \frac{1}{n+1} \right)$

10. If  $\sum a_n$  is convergent and  $\sum b_n$  is divergent, show that the series  $\sum (a_n + b_n)$  is divergent. *Hint:* Argue by first *assuming* that  $\sum (a_n + b_n)$  is convergent and using the series  $\sum (-a_n)$  to help show that this assumption is impossible.

11.  $\sum_{n=1}^{\infty} \left( \frac{2}{n+1} - \frac{3}{5^n} \right)$  *Hint: Consider the work from question #10.*